AISI 4320 Steel, normalized at 895°C (1640°F)

Categories:	Metal; Ferrous Metal; Alloy Steel; AISI 4000 Series Steel; Low Alloy Steel; Carbon Steel; Low Carbon Steel
Material Notes:	Heat treatment: Direct quench for pot - carburized at 925°C (1700°F) for 8 hours, quenched in agitated oil, 150°C (300°F) temper.
Key Words:	UNS G43200, ASTM A322, ASTM A331, ASTM A505, ASTM A519, ASTM A535
Vendors:	No vendors are listed for this material. Please click here if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	Metric	English	Comments
Density	7.85 g/cc	0.284 lb/in³	

Mechanical Properties	Metric	English	Comments
Hardness, Brinell	235	235	
Hardness, Knoop	259	259	Converted from Brinell
Hardness, Rockwell B	97	97	Converted from Brinell
Hardness, Rockwell C	21	21	Converted from Brinell
Hardness, Vickers	247	247	Converted from Brinell
Tensile Strength, Ultimate	793 MPa	115000 psi	
Tensile Strength, Yield	460 MPa	66700 psi	
Elongation at Break	20.8 %	20.8 %	in 50 mm
Reduction of Area	51 %	51 %	
Modulus of Elasticity	205 GPa	29700 ksi	Typical for steel
Bulk Modulus	160 GPa	23200 ksi	Typical for steel
Poissons Ratio	0.29	0.29	Calculated
Machinability	60 %	60 %	annealed and cold drawn. Based on 100% machinability for AISI 1212 steel.
Shear Modulus	80.0 GPa	11600 ksi	Typical for steels.
Izod Impact	73.0 J	53.8 ft-lb	

Electrical Properties	Metric	English	Comments
Electrical Resistivity	0.0000245 ohm-cm	0.0000245 ohm-cm	Typical 4000 series steel

Thermal Properties	Metric	English	Comments
CTE, linear	11.3 µm/m-°C @Temperature -18.0 - 95.0 °C	6.28 µin/in-°F @Temperature -0.400 - 203 °F	condition of specimen unknown
Specific Heat Capacity	0.475 J/g-°C	0.114 BTU/lb-°F	Typical 4000 series steel
Thermal Conductivity	44.5 W/m-K	309 BTU-in/hr-ft²-°F	Typical steel

Component Elements Properties	Metric	English	Comments
Carbon, C	0.17 - 0.22 %	0.17 - 0.22 %	
Chromium, Cr	0.40 - 0.60 %	0.40 - 0.60 %	
Iron, Fe	95.855 - 96.98 %	95.855 - 96.98 %	As remainder
Manganese, Mn	0.45 - 0.65 %	0.45 - 0.65 %	
Molybdenum, Mo	0.20 - 0.30 %	0.20 - 0.30 %	
Nickel, Ni	1.65 - 2.0 %	1.65 - 2.0 %	
Phosphorus, P	<= 0.035 %	<= 0.035 %	
Silicon, Si	0.15 - 0.30 %	0.15 - 0.30 %	
Sulfur, S	<= 0.040 %	<= 0.040 %	

References for this datasheet.